EXPERIMENT – 23 :Prolog Program to Implement Family Tree

Aim:

To write and execute a Prolog program to implement a family tree using facts and rules to identify family relationships such as father, mother, parent, brother, sister, grandfather, grandmother and siblings.

Program

File Name: exp23.pl

File Path: C:/ai prolog/exp23.pl

```
% Family Tree - Facts

male(ravi).
male(kumar).
male(rajesh).
male(rahul).
male(amit).

female(priya).
female(lakshmi).
female(sita).
female(anu).

% Parent Relationships

parent(ravi, kumar).
parent(priya, kumar).

parent(ravi, sita).
parent(priya, sita).
```

```
parent(kumar, rahul).
parent(lakshmi, rahul).

parent(kumar, anu).
parent(lakshmi, anu).

parent(rajesh, amit).
parent(sita, amit).

% Rules

father(X, Y) :-
    parent(X, Y),
    male(X).

mother(X, Y) :-
    parent(X, Y),
    female(X).

brother(X, Y) :-
    parent(Z, X),
    parent(Z, Y),
    male(X),
    X \= Y.

sister(X, Y) :-
    parent(Z, X),
```

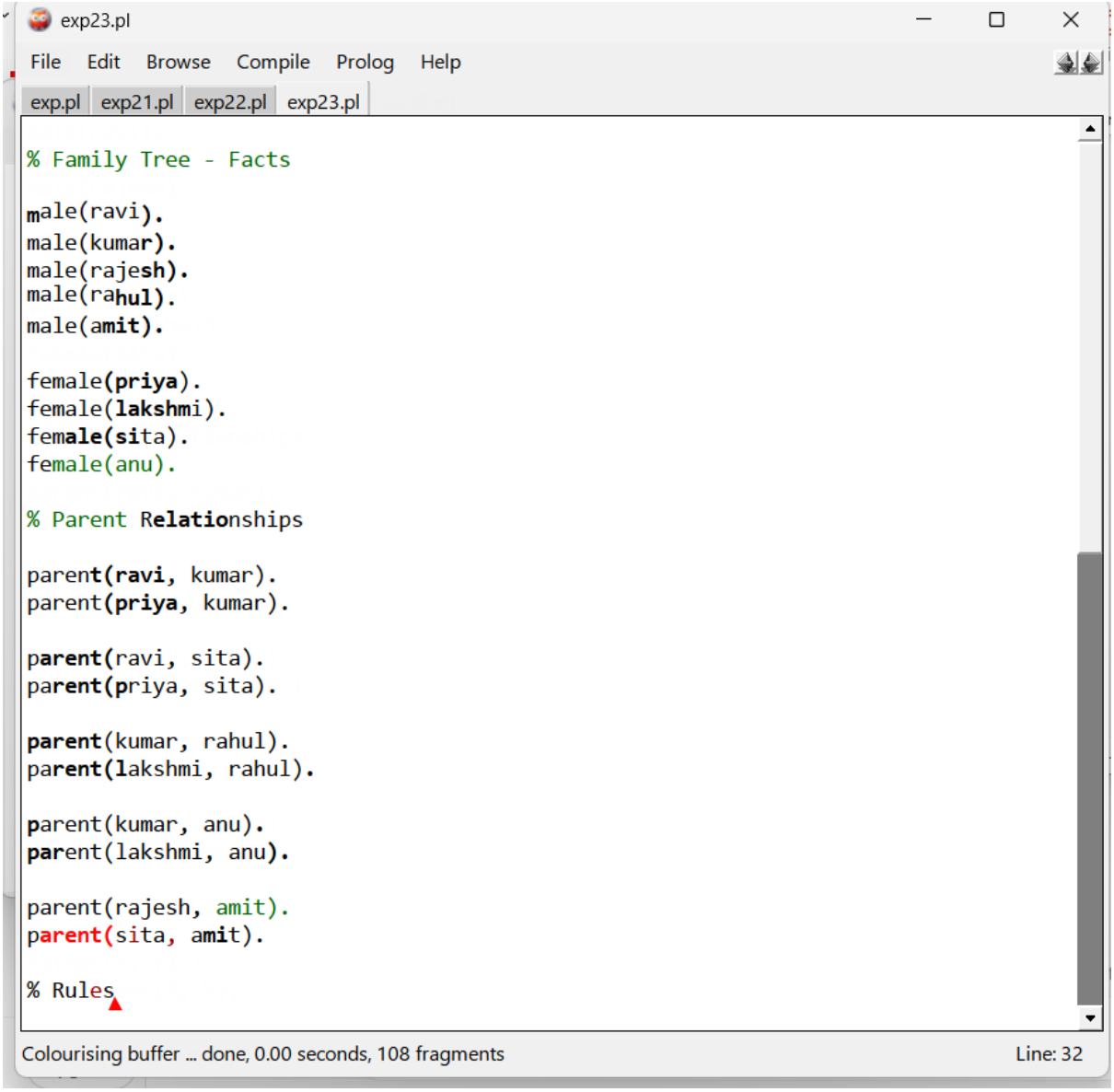
```
    parent(Z, Y),
    female(X),
    X \= Y.

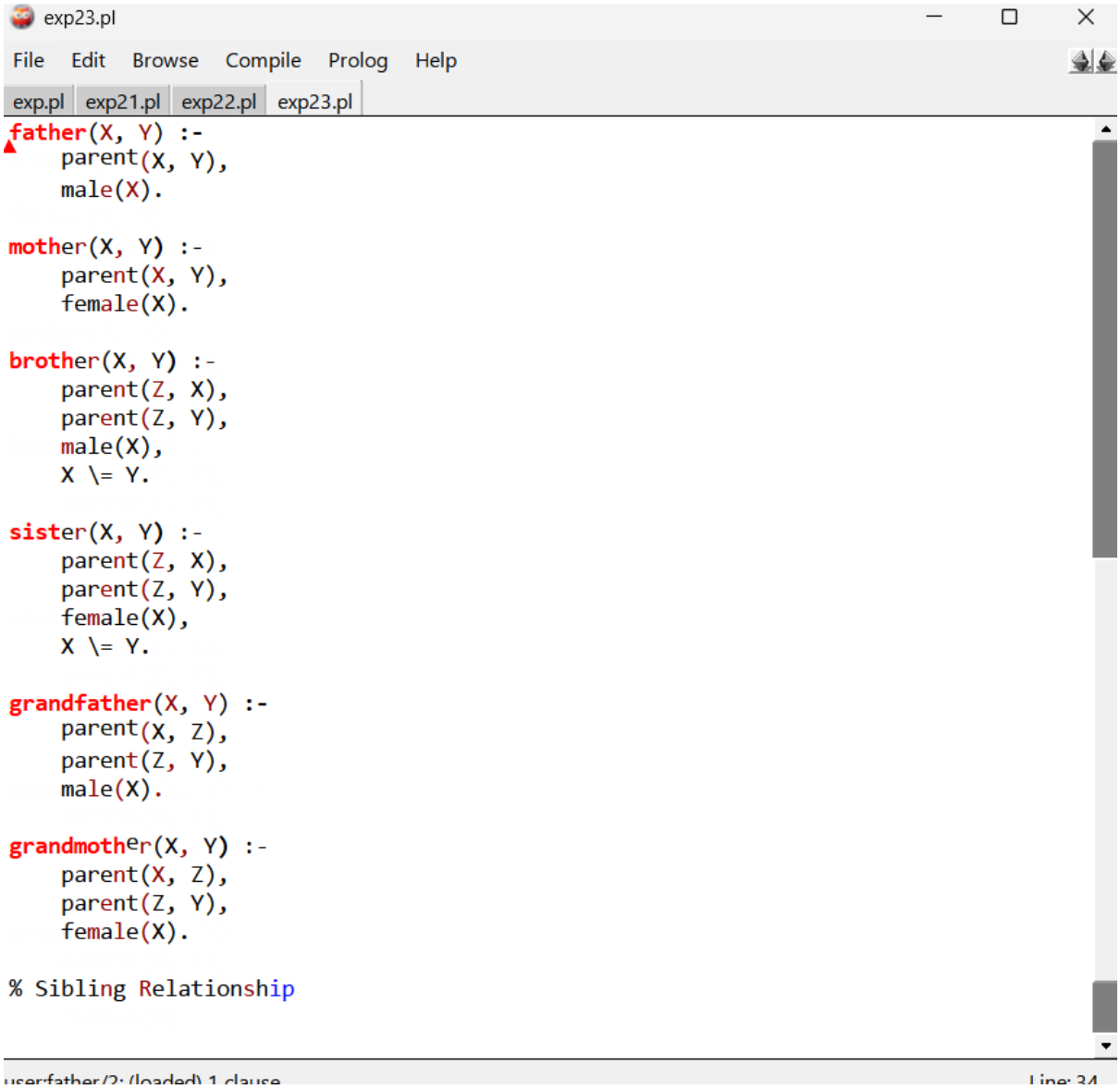
grandfather(X, Y) :-
    parent(X, Z),
    parent(Z, Y),
    male(X).

grandmother(X, Y) :-
    parent(X, Z),
    parent(Z, Y),
    female(X).

% Sibling Relationship

sibling(X, Y) :-
    parent(Z, X),
    parent(Z, Y),
    X \= Y.
```


exp23.pl
File Edit Browse Compile Prolog Help
exp.pl exp21.pl exp22.pl exp23.pl
% Family Tree - Facts
male(ravi).
male(kumar).
male(rajesh).
male(rahul).
male(amit).
female(priya).
female(lakshmi).
female(sita).
female(anu).
% Parent Relationships
parent(ravi, kumar).
parent(priya, kumar).
parent(ravi, sita).
parent(priya, sita).
parent(kumar, rahul).
parent(lakshmi, rahul).
parent(kumar, anu).
parent(lakshmi, anu).
parent(rajesh, amit).
parent(sita, amit).
% Rules
Colourising buffer ... done, 0.00 seconds, 108 fragments
Line: 32



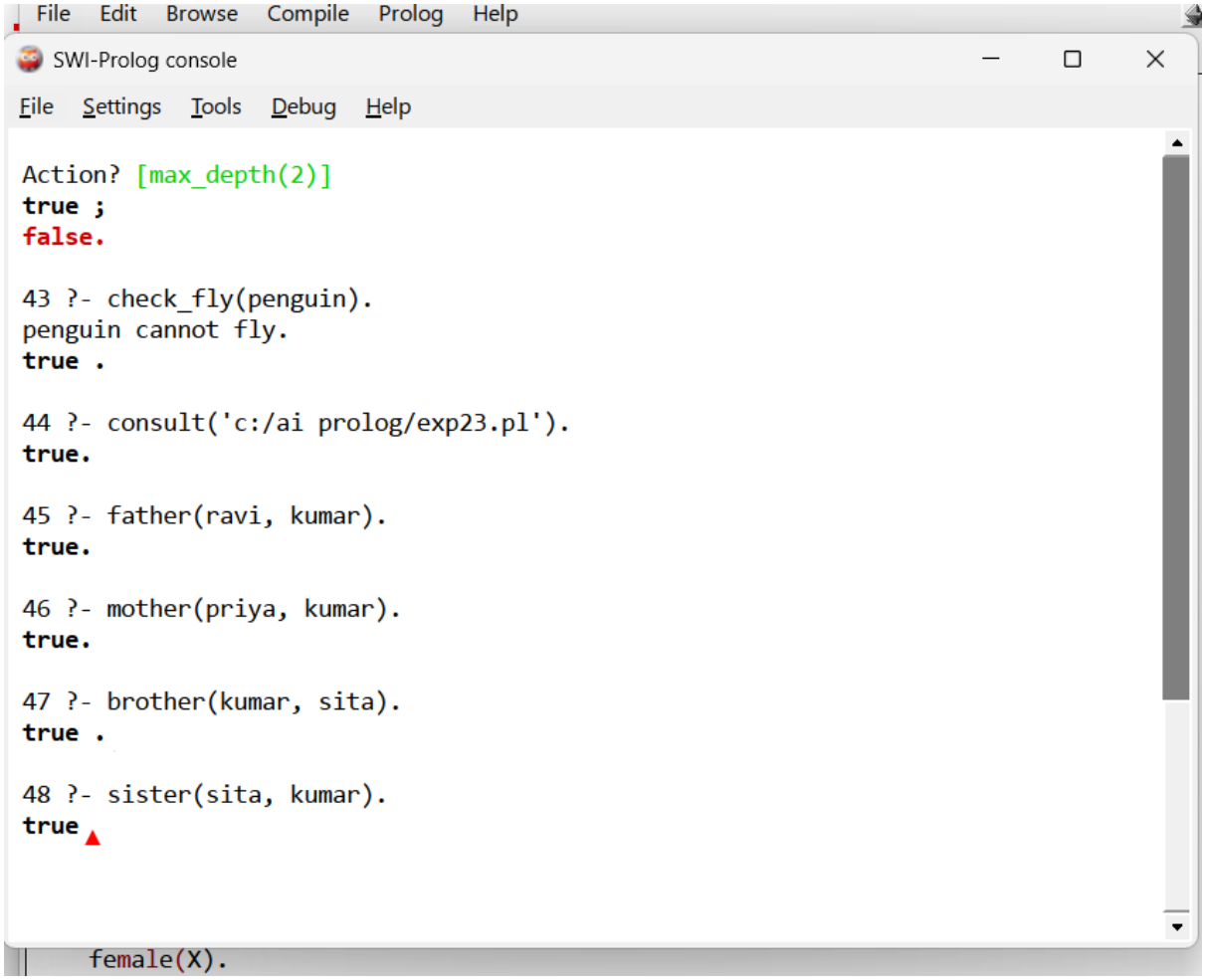
Procedure

1. Open SWI-Prolog software.
2. Create a new file and enter the above program.
3. Save the file as exp23.pl in the folder C:/ai prolog/.
4. Load the program using the following command:

?- consult('C:/ai prolog/exp23.pl').

5. Execute the required queries one by one.
6. Observe the output in the Prolog console.

Required Queries and Output



Explanation

- Facts: Store the gender and parent relationships of family members.
- Father: A male person who is a parent.
- Mother: A female person who is a parent.
- Brother: A male person who shares a parent with another person.
- Sister: A female person who shares a parent with another person.
- Grandfather: A male person who is the parent of someone's parent.
- Grandmother: A female person who is the parent of someone's parent.
- Sibling: Two different people who share at least one parent.

Result

Thus, the Prolog program to implement a family tree and identify various family relationships using facts, rules and queries was written and executed successfully.